

CALIBRATION CHALLENGE



Help Laura calibrate her sprayer before treating her vineyard.



START CHALLENGE

1

Understanding calibration

2

Setting the task

3

Measuring parameters

4

Selecting nozzles

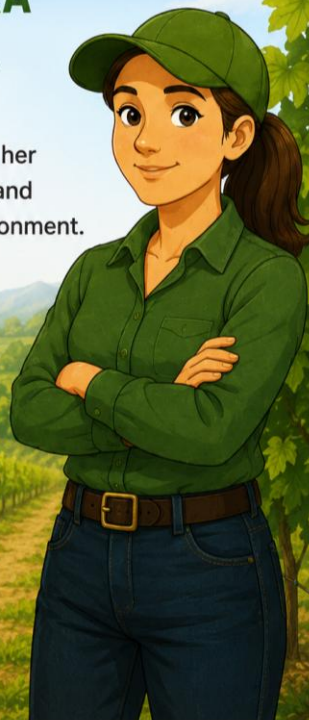
5

Verifying results

MEET LAURA

Laura is a passionate
vineyard grower.

She wants to protect her
crops, reduce waste and
take care of the environment.



VINEYARD
GROWER



CARES ABOUT
THE ENVIRONMENT



SEEKS
BEST RESULTS

MEET MIA

Mia is Laura's advisor.

She helps her make the right decisions and get the best results with her sprayer.



EXPERT
ADVISOR



PRACTICAL
GUIDANCE



ALWAYS BY
YOUR SIDE

THE SITUATION

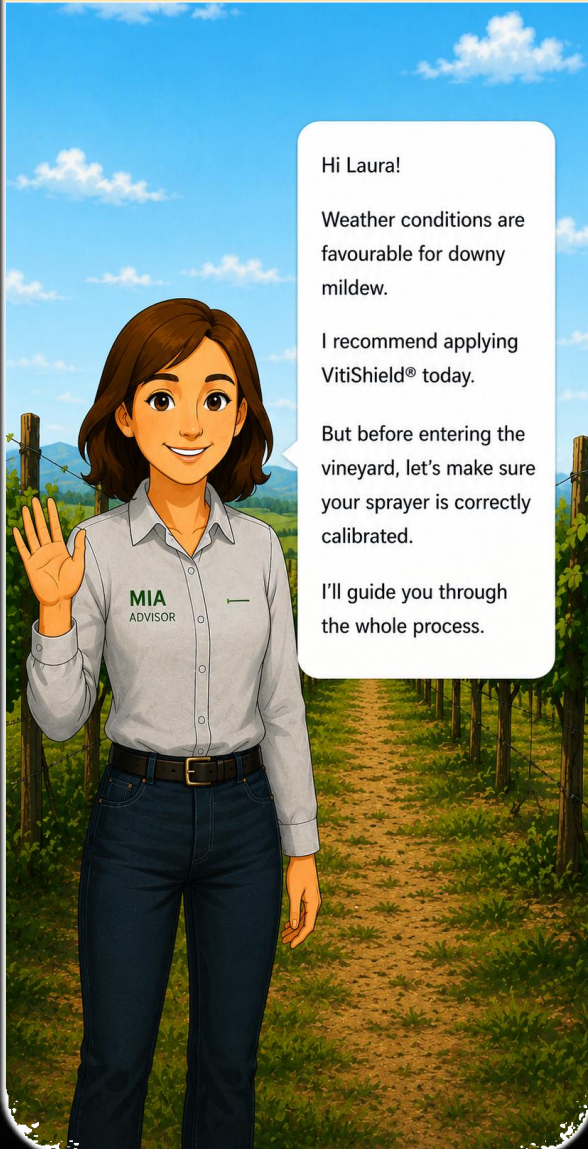
Warm and humid weather favours downy mildew infection.



Timely and well-calibrated spraying is essential.



RENOVATE APP



Hi Laura!

Weather conditions are favourable for downy mildew.

I recommend applying VitiShield® today.

But before entering the vineyard, let's make sure your sprayer is correctly calibrated.

I'll guide you through the whole process.



Weather conditions favour downy mildew infection.

Mia recommends applying VitiShield®
Fungicide at **1 L/ha** (or 100 cm³/100 L
of water), using **250 L/ha** and a
medium droplet size (M).

Before spraying, Laura must
calibrate her sprayer correctly.

 RENOVATE APP

1

UNDERSTANDING
THE CALIBRATION
PROCESS AND ITS
IMPORTANCE



MIA'S TIP

Too much spray wastes product and harms the environment. Too little won't protect your crops!

THE 3 PILLARS OF EFFECTIVENESS



CORRECT SPRAY VOLUME RATE (L/ha)

Tailored to the specific area of your vineyard.



EXACT CONCENTRATION

The right mix of the plant protection product.



PRECISE APPLICATION

Ensuring the exact dose safely reaches the target.



Calibration prepares your sprayer for efficient and safe operation.

It allows a precise and uniform application of plant protection products at the intended rate.

WHAT IS CALIBRATION?

Calibration is a procedure that:



PREPARES THE SPRAYER

for efficient and safe operation.



PERFORMS A PRECISE

application of plant protection products.



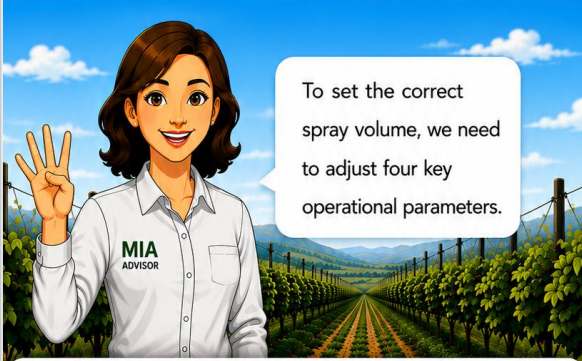
ENSURES A UNIFORM APPLICATION

at the intended application rate.



REMEMBER

Calibration is the practical and reliable way to achieve the right spray volume rate in your vineyard or orchard.



THE 4 KEY PARAMETERS

These four parameters work together to determine how much spray reaches the crop.



Changing one parameter affects the others and the final spray volume applied.

1

DRIVING SPEED
(km/h)



2

SPRAY PRESSURE
(bar)



3

NOZZLE SIZE



4

**NUMBER OF
ACTIVE NOZZLES**





CALIBRATION OF A SPRAYER

What is calibration of a sprayer about?

A

Starting up a newly purchased sprayer.

B

Inspecting the sprayer for certification purposes.

C

Setting the sprayer's operating parameters to ensure precise pesticide application.

D

Preparing the sprayer for sale.



CALIBRATION OF A SPRAYER

What is calibration of a sprayer about?

CORRECT

C

Setting the sprayer's operating parameters to ensure precise pesticide application.

i

Calibration determines the appropriate settings needed to apply the correct amount of spray accurately and uniformly.



CALIBRATION OF A SPRAYER

What is calibration of a sprayer about?

INCORRECT

A

Starting up a newly purchased sprayer.

B

Inspecting the sprayer for certification purposes.

D

Preparing the sprayer for sale.

i

Calibration is not about purchasing, inspecting or selling a sprayer. Its purpose is to adjust the sprayer so it delivers the intended application rate under real working conditions.



CALIBRATION OF A SPRAYER

Which parameters determine the spray application rate?

- A** Total nozzle flow rate and forward speed.
- B** Pump speed and agitation.
- C** The spray liquid volume in relation to the tank capacity.
- D** Nozzle size pressure, forward speed, and number of active nozzles.



CALIBRATION OF A SPRAYER

Which parameters determine the spray application rate?

CORRECT

A

Total nozzle flow rate and forward speed.

D

Nozzle size pressure, forward speed, and number of active nozzles.

i

Spray application rate depends on nozzle flow rate, operating pressure, forward speed, and the number of active nozzles.



CALIBRATION OF A SPRAYER

Which parameters determine the spray application rate?

INCORRECT

B

Pump speed and agitation.

C

The spray liquid volume in relation to the tank capacity.

i

Tank filling level, agitation or pump speed are important for proper sprayer operation, but they do not determine the application rate per hectare.



CALIBRATION OF A SPRAYER

What happens when spray pressure increases?

A

Reduced pressure pulsation.

B

Higher nozzle flow rate.

C

Increased nozzle durability.

D

Finer droplets produced.



CALIBRATION OF A SPRAYER

What happens when spray pressure increases?

CORRECT

B

Higher nozzle flow rate.

D

Finer droplets produced.

i

Higher pressure increases nozzle output and generally produces finer droplets.



CALIBRATION OF A SPRAYER

What happens when spray pressure increases?

INCORRECT

A

Reduced pressure pulsation.

C

Increased nozzle durability.

i

Increasing pressure does not improve nozzle durability or eliminate pressure pulsations.



CALIBRATION OF A SPRAYER

If nozzle flow rate and pressure remain unchanged, why does increasing forward speed reduce spray quality?



- A Lower application rate.
- B Changes pressure and droplet size.
- C Reduced canopy penetration.
- D It has no negative effect because pressure automatically compensates.



CALIBRATION OF A SPRAYER

If nozzle flow rate and pressure remain unchanged, why does increasing forward speed reduce spray quality?



CORRECT

A

Lower application rate.

C

Reduced canopy penetration.



Increasing speed reduces the spray volume applied per hectare and the time available for canopy penetration.



CALIBRATION OF A SPRAYER

If nozzle flow rate and pressure remain unchanged, why does increasing forward speed reduce spray quality?



INCORRECT

B Changes pressure and droplet size.

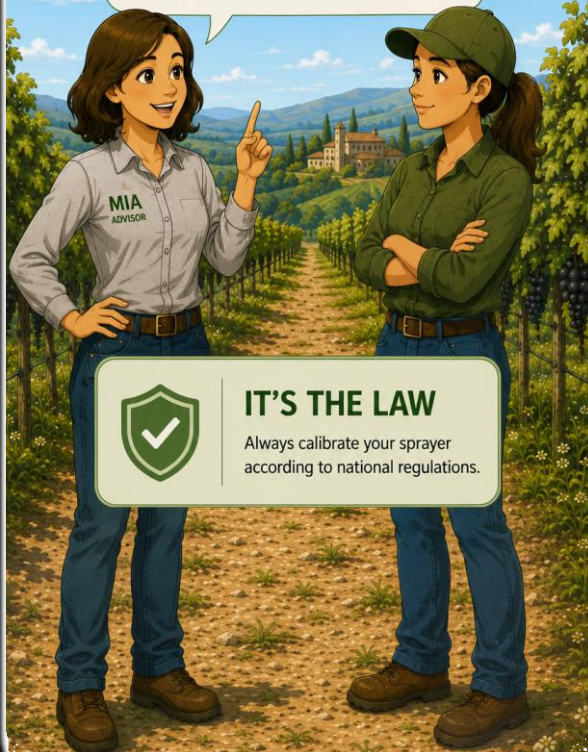
D It has no negative effect because pressure automatically compensates.

i Driving faster does not automatically increase pressure or change droplet size enough to compensate for the higher speed.



**Calibration
is not just a
good practice –
it is a legal
requirement.**

It arises from national regulations
and is essential for safe and
responsible plant protection.



IT'S THE LAW

Always calibrate your sprayer
according to national regulations.

WHY CALIBRATION MATTERS

Calibration brings many benefits.
Not calibrating has serious consequences.



BENEFITS OF CALIBRATION



Effective pest and disease control

Ensures the intended spray volume rate is applied accurately.



Cost savings

Avoids over-application of plant protection products, water and fuel.



Environmental protection

Reduces the risk of run-off, drift and contamination.



Compliance with regulations

Helps meet legal requirements and standards.



Better performance of the sprayer

Extends equipment life and reduces the risk of technical problems.



Professionalism and knowledge

Improves the operator's skills and decision-making.



CONSEQUENCES OF NOT CALIBRATING



Poor efficacy

Under-application leads to insufficient pest and disease control.



Wastage of resources

Over-application results in unnecessary use of plant protection products, water and fuel.



Environmental harm

Higher risk of pollution, residue and negative impact on non-target organisms.



Legal and financial risks

Non-compliance can lead to penalties and loss of support.



Equipment wear and damage

Incorrect settings increase the risk of malfunction and premature wear.



Loss of reputation and profitability

Lower yields, market rejection and increased costs.



MIA'S TIP

A well-calibrated sprayer protects your crops, your money, your environment and your reputation.

Take a few minutes to calibrate – it makes a big difference!



CALIBRATION OF A SPRAYER

What are the main benefits of calibration?

A Environmental benefits only.

B Lower plant protection costs.

C Verified sprayer readiness.

D Better repeatability and documentation.



CALIBRATION OF A SPRAYER

What are the main benefits of calibration?

CORRECT

B

Lower plant protection costs.

C

Verified sprayer readiness.

D

Better repeatability and documentation.

i

Calibration improves efficiency, reliability, consistency, and reduces unnecessary costs.



CALIBRATION OF A SPRAYER

What are the main benefits of calibration?

INCORRECT

A

Environmental benefits only.



Calibration benefits both the environment and the operator by improving efficiency reducing waste and increasing application reliability.



CALIBRATION OF A SPRAYER

Which risks can proper calibration help avoid?

A

Operator contamination during filling.

B

Excess spray liquid remaining.

C

Product sedimentation.

D

Running out of spray liquid before finishing.



CALIBRATION OF A SPRAYER

Which risks can proper calibration help avoid?

CORRECT

B

Excess spray liquid remaining.

D

Running out of spray liquid before finishing.

i

Proper calibration matches spray volume to the treated area, minimising leftovers and premature tank emptying.



CALIBRATION OF A SPRAYER

Which risks can proper calibration help avoid?

INCORRECT

A

Operator contamination during filling.

C

Product sedimentation.

i

Calibration does not prevent operator exposure or product sedimentation. Its purpose is to ensure the correct spray volume, helping reduce leftover spray mixture and preventing the sprayer from running out before the treatment is completed.



CALIBRATION OF A SPRAYER

What are the main consequences of not calibrating?

A

Increased spray drift and environmental contamination.

B

More frequent inspections.

C

No consequences.

D

Possible legal penalties.



CALIBRATION OF A SPRAYER

What are the main consequences of not calibrating?

CORRECT

A

Increased spray drift and environmental contamination.

D

Possible legal penalties.



Poor calibration increases environmental risk and may lead to regulatory non-compliance.



CALIBRATION OF A SPRAYER

What are the main consequences of not calibrating?

INCORRECT

B

More frequent inspections.

C

No consequences.

i

Mandatory inspections cannot replace calibration, and inspection frequency does not depend on whether the sprayer has been calibrated.

Hi Laura!

Before we start calibrating the sprayer, remember the most important rule:

calibration must always be safe.

SAFETY FIRST

Calibration must protect both:



PEOPLE



ENVIRONMENT



MIA'S TIP

Even when using clean water,
always work safely.

RENOVATE APP

Even after cleaning,
a sprayer may still contain
pesticide residues.
These residues can lead to
exposure to chemicals.



CHEMICAL RESIDUES

Pesticide residues may remain inside the tank, hoses and nozzles, even after cleaning.



SKIN CONTACT

Contact with residues can be absorbed through the skin.



SPLASH RISK

During calibration, splashes or leaks can cause exposure to chemicals.



ALWAYS USE PPE

Personal Protective Equipment (PPE)

Protect yourself and prevent contamination of people and the environment.



MIA'S TIP

Using basic personal protective equipment is essential during calibration.
It's a simple step that makes a big difference.



BASIC PPE FOR CALIBRATION



FACE SHIELD

Protects eyes, nose and mouth from splashing.



COVERALL

Certified for protection against chemical substances.



GLOVES

Certified for use with pesticides.



MIA'S TIP

PPE is your first line of defence.
Take a few minutes to wear it properly –
your health is the most important!



CALIBRATION OF A SPRAYER

How should the sprayer be prepared for calibration?

A Sprayer tank completely filled.

B Fill the tank halfway with clean water.

C Add pesticide.

D Clean the sprayer.



CALIBRATION OF A SPRAYER

How should the sprayer be prepared for calibration?

CORRECT

B

Fill the tank halfway with clean water.

D

Clean the sprayer.

i

Calibration should be performed using a clean sprayer partially filled with clean water.



CALIBRATION OF A SPRAYER

How should the sprayer be prepared for calibration?

INCORRECT

A

Sprayer tank completely filled.

C

Add pesticide.

i

Calibration should always be performed with clean water, and a partially filled tank provides enough clean water for calibration without unnecessary weight or waste.



CALIBRATION OF A SPRAYER

What type of protection would you use to safely perform sprayer calibration?

A

Ordinary rubber gloves.

B

Chemical resistant pesticide gloves.

C

Waterproof outdoor jacket.

D

Helmet.



CALIBRATION OF A SPRAYER

What type of protection would you use to safely perform sprayer calibration?

CORRECT

B Chemical-resistant pesticide gloves.



Chemical-resistant gloves provide appropriate protection against pesticide exposure.



CALIBRATION OF A SPRAYER

What type of protection would you use to safely perform sprayer calibration?

INCORRECT

A

Ordinary rubber gloves.

C

Waterproof outdoor jacket.

D

Helmet.



Ordinary gloves, outdoor clothing or helmets are not designed to protect against pesticide contamination. Certified chemical-resistant gloves are essential to prevent exposure to pesticide residues that may still be present inside the sprayer.



CALIBRATION OF A SPRAYER

Which PPE should be worn during calibration?

A

Cotton apron.

B

Chemical resistant coverall.

C

Respirator.

D

Face shield.



CALIBRATION OF A SPRAYER

Which PPE should be worn during calibration?

CORRECT

B

Chemical-resistant coverall.

D

Face shield.

i

Chemical-resistant clothing and face protection reduce exposure during calibration.



CALIBRATION OF A SPRAYER

Which PPE should be worn during calibration?

INCORRECT

A

Cotton apron.

C

Respirator.

i Not all protective equipment provides adequate protection during sprayer calibration. Always use PPE specifically designed for pesticide handling to protect against contaminated splashes and residues.

 RENOVATE APP

2

SETTING THE TASK
BY DEFINING THE
TARGET SPRAY
VOLUME

WHY DO WE NEED TO DEFINE THE SPRAY VOLUME?

Mia, how do I know how much spray I should apply?

The spray volume depends on vineyard characteristics. For this vineyard, the recommended application rate is 250 L/ha.



SPRAY VOLUME DEPENDS ON

Canopy volume



Canopy density



WHAT HAPPENS IF I APPLY MORE OR LESS THAN THE RECOMMENDED VOLUME?

Mia, what happens if I apply more or less than the recommended volume?

Good question, Laura! Applying more or less than the recommended volume can reduce the effectiveness of the treatment and increase costs or risks.

+ APPLYING MORE THAN RECOMMENDED



HIGHER COSTS

You use more product than necessary, which increases the cost of the treatment.



GREATER RISK OF RUNOFF AND LEACHING

More product can be lost to the environment through runoff or leaching, contaminating water and harming ecosystems.



SPRAY RETENTION RISK

An excessively high volume of diluted spray liquid reduces liquid retention on leaves, consequently decreasing the deposition and effectiveness of the active substance.



NO INCREASE IN EFFICACY

More volume does not mean better control. There is a limit beyond which effectiveness does not improve.

- APPLYING LESS THAN RECOMMENDED



POOR COVERAGE

The product may not reach all the target area, leaving parts of the canopy untreated.



LOWER TREATMENT EFFICACY

Pests or diseases may not be well controlled, leading to damage and yield losses.



RISK OF RESISTANCE

Repeated poor control can allow pests or diseases to survive and develop resistance to the product.



WASTED EFFORT

Time, fuel and labor are used, but the results are not achieved, requiring additional treatments.



SPRAY VOLUME

What is spray volume rate?

- A** Amount of spray liquid applied per unit ground area (L/ha).
- B** Nozzle flow rate (L/min).
- C** Total sprayer flow rate (L/min).
- D** Spray tank volume (L).



SPRAY VOLUME

What is spray volume rate?

CORRECT

A

Amount of spray liquid applied per unit ground area (L/ha).

i

Spray volume rate expresses the amount of spray applied per hectare.



SPRAY VOLUME

What is spray volume rate?

INCORRECT

B

Nozzle flow rate (L/min).

C

Total sprayer flow rate (L/min).

D

Spray tank volume (L).



Spray volume rate is not the sprayer's flow rate or tank capacity. It describes the amount of spray liquid applied per hectare, allowing the correct dose to be distributed over the crop.



SPRAY VOLUME

What is the significance of spray volume rate?

- A** Determines canopy coverage and spray distribution.
- B** Determines tank agitation.
- C** Influences refill frequency and field capacity.
- D** Determines pesticide concentration for a given dose.



SPRAY VOLUME

What is the significance of spray volume rate?

CORRECT

- A** Determines canopy coverage and spray distribution.
- C** Influences refill frequency and field capacity.
- D** Determines pesticide concentration for a given dose.

i Spray volume rate affects coverage, efficiency, refilling frequency, and spray concentration.



SPRAY VOLUME

What is the significance of spray volume rate?

INCORRECT

B

Determines tank agitation.



Spray volume rate is not related to the sprayer's hydraulic system. Its purpose is to ensure adequate canopy coverage while maintaining efficient spraying and the correct pesticide concentration.



SPRAY VOLUME

What determines the required spray volume rate?

A

Tree canopy dimensions.

B

Sprayer tank capacity.

C

Nozzle type.

D

Tree row spacing.



SPRAY VOLUME

What determines the required
spray volume rate?

CORRECT

A

Tree canopy dimensions.



Spray volume is selected according to
canopy size and orchard geometry.



SPRAY VOLUME

What determines the required spray volume rate?

INCORRECT

B

Sprayer tank capacity.

C

Nozzle type.

D

Tree row spacing.

i

Spray volume is not determined by the sprayer or the nozzles themselves. It is selected according to canopy dimensions and tree row spacing of the sprayer to match the amount of vegetation being treated.



SPRAY VOLUME

Why match spray volume rate to the canopy?

A

Prevent spray runoff.

B

Improve pruning.

C

Improve flowering.

D

Ensure effective spray coverage.



SPRAY VOLUME

Why match spray volume rate to the canopy?

CORRECT

A

Prevent spray runoff.

D

Ensure effective spray coverage.

i

Appropriate spray volume maximizes coverage while minimising runoff.



SPRAY VOLUME

Why match spray volume rate to the canopy?

INCORRECT

B

Improve pruning.

C

Improve flowering.



Matching the spray volume to the canopy is not intended to influence crop growth or orchard management. Its purpose is to achieve adequate canopy coverage while avoiding excessive runoff, ensuring efficient and effective pesticide application.



SPRAY VOLUME

Consequences of excessive spray volume?

A

Spray runoff.

B

Longer spraying time.

C

Wasted water.

D

Better deposition without losses.



SPRAY VOLUME

Consequences of excessive spray volume?

CORRECT

A

Spray runoff.

B

Longer spraying time.

C

Wasted water.

i

Excessive spray volume increases runoff, refill frequency, and water consumption.



SPRAY VOLUME

Consequences of excessive spray volume?

INCORRECT

D Better deposition without losses.

i More spray does not always mean better results. Once the canopy reaches its retention capacity, additional spray runs off the leaves, increasing losses without improving coverage.



SPRAY VOLUME

Which consequences do you expect from an insufficient spray volume?

A

More frequent refilling.

B

Reduced efficacy.

C

Uneven coverage.

D

Smaller fruit.



SPRAY VOLUME

Which consequences do you expect from an insufficient spray volume?

CORRECT

B

Reduced efficacy.

C

Uneven coverage.

i

Too little spray reduces target coverage and treatment efficacy.



SPRAY VOLUME

Which consequences do you expect from an insufficient spray volume?

INCORRECT

A

More frequent refilling.

D

Smaller fruit.

i

A low spray volume mainly affects spray coverage, not crop growth or sprayer productivity. Insufficient coverage may leave pests or diseases insufficiently controlled.

 RENOVATE APP

3

MEASURING
PARAMETERS

LET'S CALIBRATE THE SPRAYER



Great! Now that we know why calibration is important, let's proceed to **calibrate the sprayer** step by step.



Perfect,
I'm ready!



MIA'S TIP

Calibration ensures the right amount of spray is applied to your vineyard, improving effectiveness and avoiding waste.

 RENOVATE APP

3.1

**SELECT NUMBER OF
NOZZLES**

?

CHOOSE RIGHT NUMBER OF NOZZLES

Laura, how many nozzles do you plan to use? This is one of the key parameters needed for sprayer calibration?



A

8 nozzles.

B

I need to check the canopy first.

C

Always 6 nozzles.

D

All nozzles on the sprayer.



CHOOSE RIGHT NUMBER OF NOZZLES

Laura, how many nozzles do you
plan to use?



CORRECT

B

I need to check the canopy first.

i

The number of active nozzles depends on the canopy size. Only the nozzles that spray the vegetation should remain active. This improves spray efficiency and reduces losses.



CHOOSE RIGHT NUMBER OF NOZZLES

Laura, how many nozzles do you
plan to use?

INCORRECT

A

8 nozzles.

C

Always 6 nozzles.

D

All nozzles on the sprayer.

i The number of nozzles should not be
chosen arbitrarily or assumed to be
always the same. The number of active
nozzles depends on the canopy size
and shape. Before calibration, you
should inspect the vegetation and
activate only the nozzles that spray the
target.



CHOOSE RIGHT NUMBER OF NOZZLES

Why is selection of active nozzle
the first step in sprayer
calibration?

A

Increases application rate.

B

Match the spray pattern to the
canopy.

C

Reduce filling time.

D

Increase spray pressure.



CHOOSE RIGHT NUMBER OF NOZZLES

Why is selection of active nozzle
the first step in sprayer
calibration?

CORRECT

B

Match the spray pattern to the
canopy.

i

The active nozzles should match the
canopy dimensions. Spraying outside
the canopy wastes product and
increases environmental losses.



CHOOSE RIGHT NUMBER OF NOZZLES

Why is selection of active nozzles
the first step in sprayer
calibration?

INCORRECT

A

Increases application rate.

C

Reduce filling time.

D

Increase spray pressure.

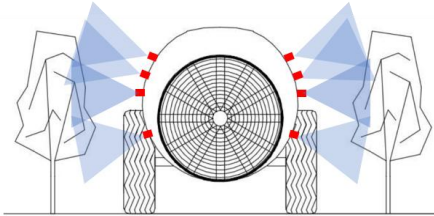
i

The purpose of nozzle selection is to adapt the spray pattern to the crop canopy. Before calculating flow rate, pressure, or speed, you must decide which nozzles should be active to ensure the spray reaches the target.

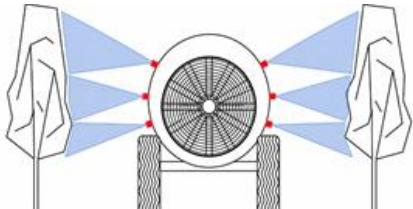


CHOOSE RIGHT NUMBER OF NOZZLES

Option A



Option B



A

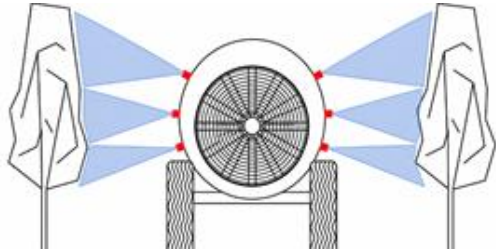
4 nozzles per side.

B

3 nozzles per side.



CHOOSE RIGHT NUMBER OF NOZZLES



CORRECT

B

Three nozzles per side.
Six nozzles in total!!!

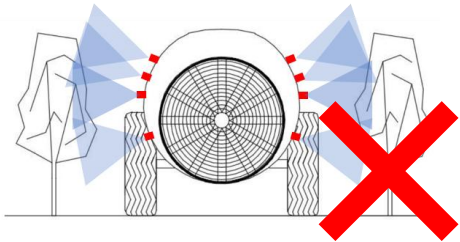
i

Only the nozzles that canopy directly
onto the canopy should be activated.

Using 3 nozzles per side directs the
spray to the vegetation avoiding losses.



CHOOSE RIGHT NUMBER OF NOZZLES



INCORRECT

A

4 nozzles per side.

i

Using **four** nozzles per side causes part of the spray to miss the canopy. Spraying above or below the target wastes product and increases environmental contamination. Always activate only the nozzles that cover the vegetation.

 RENOVATE APP

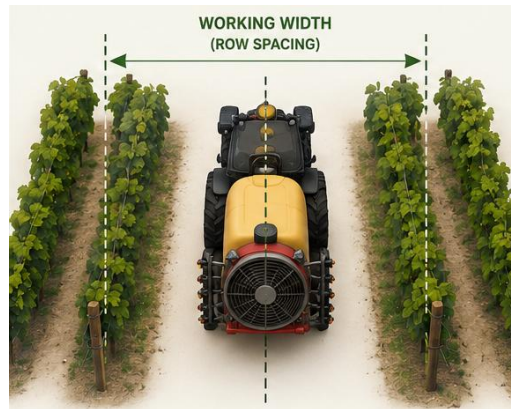
3.2

**DETERMINING
SPRAYER'S WORKING
WIDTH**

DETERMINING SPRAYER'S WORKING WIDTH



Laura, we already know the spray volume and the active nozzles. **Now we need the working width.**



i In vineyards, the working width is the distance between rows (row spacing) multiplied by the number of sprayed rows in one pass. In this case 1 row (two half rows).

DETERMINING SPRAYER'S WORKING WIDTH

- 1 Go to the field
with the sprayer.



- 2 Measure the distance
between the centers
of two rows.

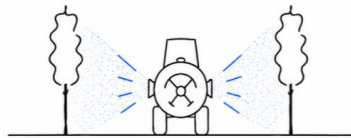


i Row spacing at Laura's
vineyard is 2.80 m.



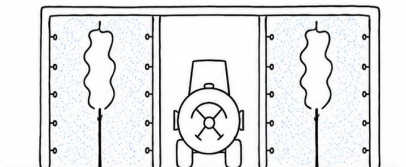
DETERMINING SPRAYER'S WORKING WIDTH

The row spacing in these vineyards is 2.8 m. **What is the working width of each sprayer?**



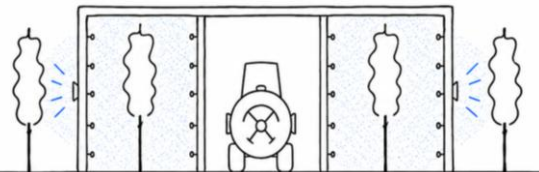
A

Conventional airblast sprayer



B

Tunnel sprayer



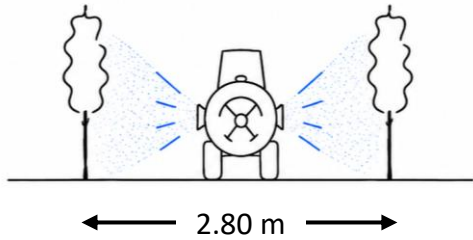
C

Multi-row sprayer



DETERMINING SPRAYER'S WORKING WIDTH

Conventional airblast sprayer



Choose the correct working
width?

A 1.4 m.

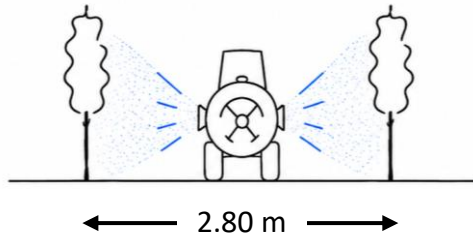
B 2.8 m.

C 5.6 m.

D 8.4 m.



DETERMINING SPRAYER'S WORKING WIDTH



Conventional airblast sprayer

CORRECT

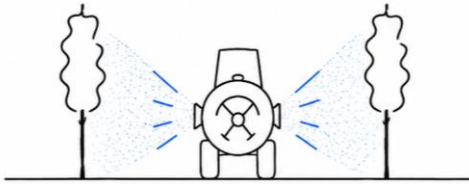
B

Working width: 2.8 m

i A conventional airblast sprayer must pass through every row alley because it sprays only **half a row on each side**. Therefore, the treated area corresponds to **1 complete row (0.5 + 0.5)** and the **working width is equal to the row spacing: 2.8 m**.



DETERMINING SPRAYER'S WORKING WIDTH



← 2.80 m →

Conventional airblast sprayer

INCORRECT

A

1.4 m.

C

5.6 m.

D

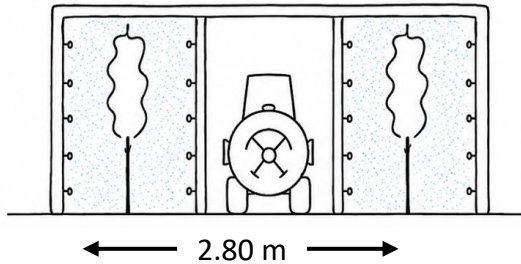
8.4 m.

i A conventional airblast sprayer sprays only **half a row on each side**. Therefore, the treated area corresponds to **1 complete row (0.5 + 0.5)** and the **working width is equal to the row spacing: 2.80 m.**



DETERMINING SPRAYER'S WORKING WIDTH

Tunnel sprayer



Choose the correct working
width?

A 5.6 m.

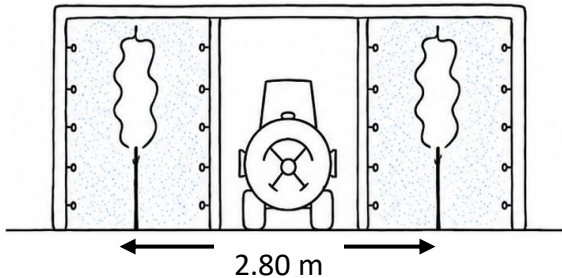
B 8.4 m.

C 11.2 m.

D 14.0 m.



DETERMINING SPRAYER'S WORKING WIDTH



Tunnel sprayer

CORRECT

A

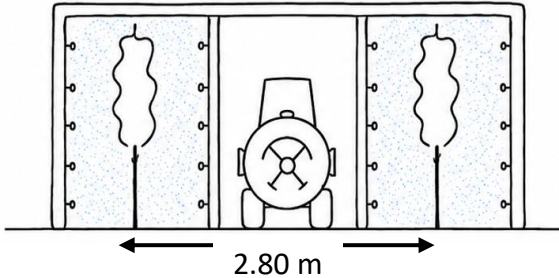
Working width: 5.6 m

i

A tunnel sprayer treats **two complete rows simultaneously**, one on each side of the machine. Therefore, the number of treated rows is **2** and the working width is: **$2 \times 2.80 \text{ m} = 5.60 \text{ m}$** .



DETERMINING SPRAYER'S WORKING WIDTH



Tunnel sprayer

INCORRECT

B 8.4 m.

C 11.2 m.

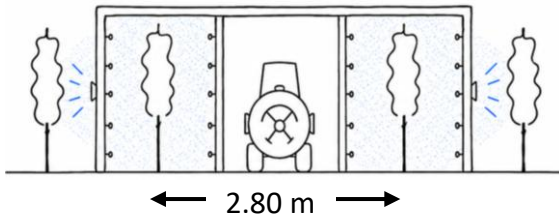
D 14.0 m.

i The working width depends on how many rows are treated during each pass. Unlike a conventional airblast sprayer, a tunnel sprayer treats two rows simultaneously, so the working width is: $2 \times 2.80 \text{ m} = 5.60 \text{ m}$.



DETERMINING SPRAYER'S WORKING WIDTH

Multi-row sprayer



Choose the correct working
width?

A 5.6 m.

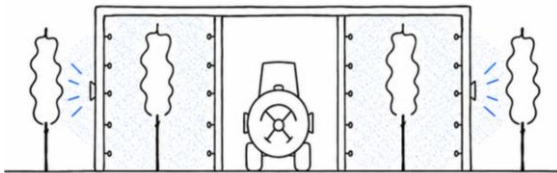
B 8.4 m.

C 11.2 m.

D 14.0 m.



DETERMINING SPRAYER'S WORKING WIDTH



← 2.80 m →
Multi-row sprayer

CORRECT

B

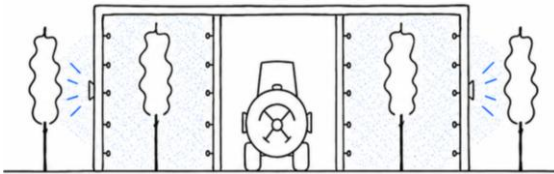
Working width: 8.4 m

i

The multi-row sprayer treats **3 rows in a single pass**: 2 complete rows + 0.5 row + 0.5 row. Therefore: **Working width = $3 \times 2.80 \text{ m} = 8.40 \text{ m}$.**



DETERMINING SPRAYER'S WORKING WIDTH



← 2.80 m →

Multi-row sprayer

INCORRECT

A

5.6 m.

C

11.2 m.

D

14.0 m.

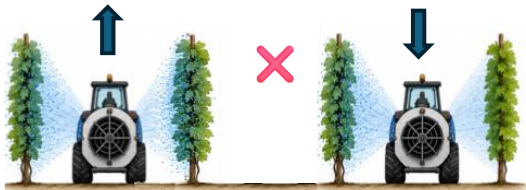


The working width depends on the number of rows treated during each pass. To calculate it, multiply the row sprayed simultaneously. The multi-row sprayer treats **3 rows in a single pass**: 2 complete rows + 0.5 + 0.5 row. Therefore: **Working width = 3 x 2.80 m = 8.40 m.**



SPRAYING EVERY OTHER ROW

Which are the main consequences of spraying every other row with a conventional orchard sprayer?



A

More uniform coverage.

B

High losses on soil.

C

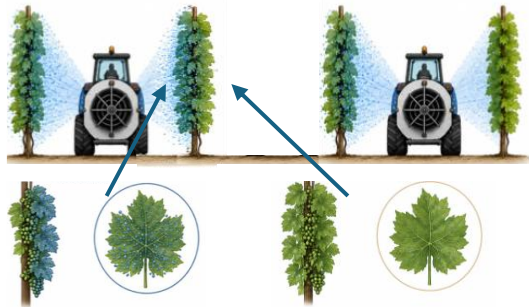
Uneven distribution.

D

Reduced penetration.



SPRAYING EVERY OTHER ROW



CORRECT

C

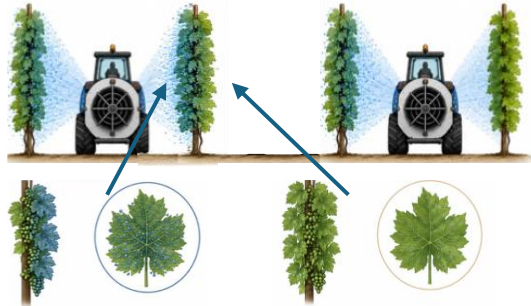
Uneven distribution.

i

Spraying every other row with a conventional airblast sprayer usually produces an uneven distribution of the spray mixture. One canopy side receives excessive deposition, while the opposite side receives insufficient coverage. Excess product may also drip from the vegetation and reach the soil.



SPRAYING EVERY OTHER ROW



INCORRECT

A

More uniform coverage.

B

High losses on soil.

D

Reduced penetration.



Spraying every other row increases the distance between the sprayer and part of the target canopy. As a result, some areas may receive too much spray while others receive too little, leading to uneven distribution.

 RENOVATE APP

3.3

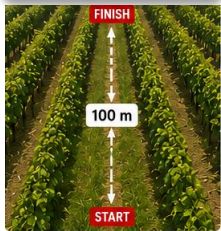
**SELECT & CHECK
FORWARD SPEED**

SELECT & CHECK FORWARD SPEED



Laura, **how fast** will you drive during the treatment? Let's check it out by following these steps!

1. Mark 100 m along the row



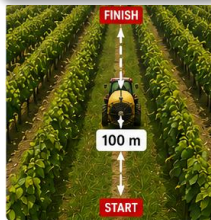
2. Select suitable gear & PTO Speed



3. Drive as normal spray conditions

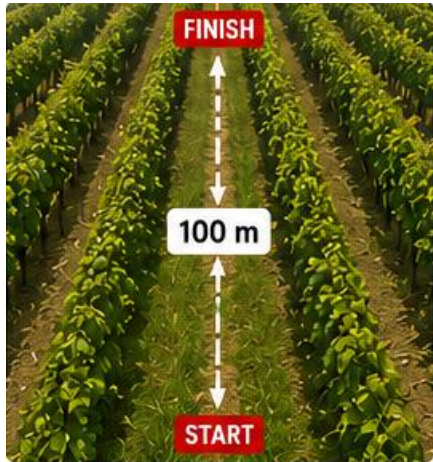


4. Check time to travel 100 m



RENOVATE APP

SELECT & CHECK FORWARD SPEED



FORMULA

$$\text{Speed (km/h)} = \frac{100 \text{ m} \times 3.6}{\text{Time (s)}}$$



SELECT & CHECK FORWARD SPEED

Laura drove on her tractor 100 m
in 72 seconds. What was her
travel speed (km/h)?

FORMULA

$$\text{Speed (km/h)} = \frac{100 \text{ m} \times 3.6}{\text{Time (s)}}$$

A

8 km/h.

B

4 km/h.

C

6 km/h.

D

5 km/h.



SELECT & CHECK FORWARD SPEED

Laura drone on her tractor 100 m
in 72 seconds. What was her
travel speed (km/h)?

CORRECT

D

5 km/h.

i

Travel speed is calculated using the
formula: travelled distance (100m) x 3.6
/ time (72 seconds) = 5.0 km/h.



SELECT & CHECK FORWARD SPEED

Laura drone on her tractor 100 m
in 72 seconds. What was her
travel speed (km/h)?

INCORRECT

A

8 km/h.

B

4 km/h.

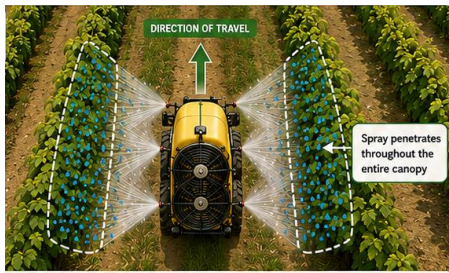
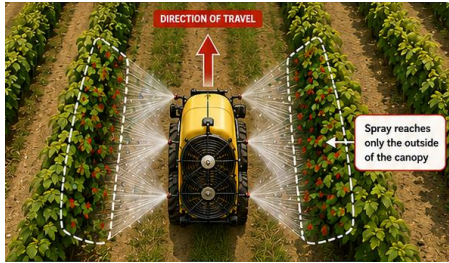
C

6 km/h.

i

Travel speed is calculated using the
formula: travelled distance (100m) x 3.6
/ time (72 seconds) = 5.0 km/h.

SELECT & CHECK FORWARD SPEED

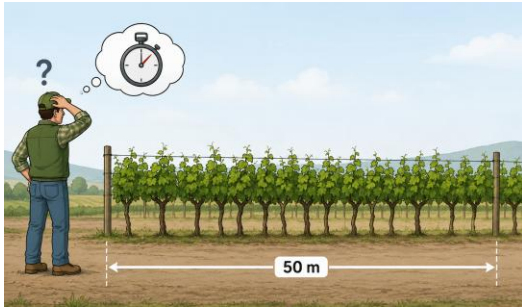


Mia's tip:

- i** Forward speed affects spray coverage and canopy penetration. As speed increases, higher airflow is required to transport droplets into the canopy. For most vineyard applications, 5-6 km/h is recommended.



SELECT & CHECK FORWARD SPEED



Laura's neighbour wants to spray his vineyard at a forward speed of 6 km/h. To check his speed, he marks a distance of 50 m in the vineyard row. Approximately how long should it take him to travel this distance?

A

20 sec.

B

30 sec.

C

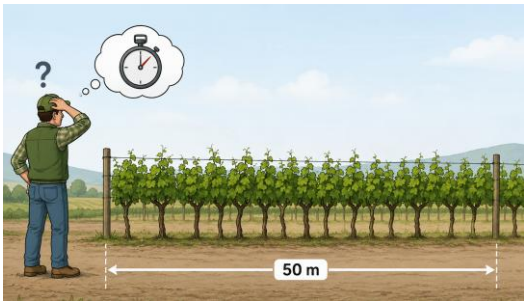
40 sec.

D

60 sec.



**SELECT & CHECK FORWARD
SPEED**



CORRECT

D

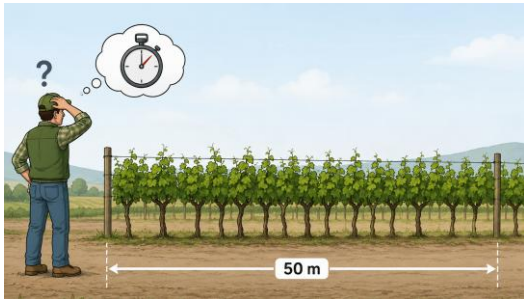
Time: 30 seconds.



A speed of **6 km/h** is equivalent to approximately **1.67 m/s**. At this speed, traveling **50 m** should take about **30 seconds**.



SELECT & CHECK FORWARD SPEED



INCORRECT

A 20 sec.

C 40 sec.

D 60 sec.

i Travel speed should always be checked under field conditions. At **6 km/h**, a tractor needs approximately **30 seconds** to travel **50 m**. An incorrect speed will lead to an incorrect application rate, even if the sprayer is properly calibrated.

 RENOVATE APP

3.4

**CALCULATE TOTAL
FLOW RATE**

CALCULATE TOTAL FLOW RATE



Laura, we are ready to calculate the **total flow rate of your sprayer!!!**

Remember: your target spray volume is 250 L/ha, you will be traveling at 5 km/h, and your row spacing is 2.8 m.

First Step: Calculate the total flow rate in L/min.

$$Q_{\text{total}} = \frac{\text{Volume} \times \text{Speed} \times \text{Working Width}}{600}$$

Using our data

$$Q_{\text{total}} = \frac{250 \times 5.0 \times 2.8}{600} = 5.83 \text{ L/min}$$

Laura needs to adjust her sprayer to deliver a total of **5.83 L/min**.



CALCULATE TOTAL FLOW RATE

What does the number 600 represent in the flow rate formula?

$$Q_{\text{total}} = \frac{\text{Volume} \times \text{Speed} \times \text{Working Width}}{600}$$
A red arrow points to the number 600 in the denominator of the formula.

- A Sprayer tank capacity.
- B PTO speed.
- C Conversion factor.
- D Number of nozzles.



CALCULATE TOTAL FLOW RATE

What does the number 600 represent in the flow rate formula?

CORRECT

C

Conversion factor.

i

The number **600** converts:

- **L/ha** for application volume,
- **km/h** for forward speed,
- **m** for working width.

Into **L/min** expressing the resultant flow rate.



CALCULATE TOTAL FLOW RATE

What does the number 600 represent in the flow rate formula?

INCORRECT

A

Sprayer tank capacity.

B

PTO speed.

D

Number of nozzles.



The number **600** is not related to the sprayer settings. It is a **unit conversion factor** that converts the application volume (L/ha), forward speed (km/h), and working width (m) into the total flow rate in **L/min**.

BE CAREFUL WITH UNITS

i Always express each parameter in the correct unit to ensure accurate and effective calibration.



APPLICATION VOLUME

Indicates the amount of spray solution applied per hectare.



L/ha

Liters
per hectare

Example: **250 L/ha**

BE CAREFUL WITH UNITS



Always express each parameter in the correct unit to ensure accurate and effective calibration.



FORWARD SPEED

Indicates the speed at which the sprayer is moving.



km/h

Kilometers
per hour

Example: **5.0 km/h**

BE CAREFUL WITH UNITS

i Always express each parameter in the correct unit to ensure accurate and effective calibration.



WORKING WIDTH

Indicates the effective width covered by the application.



m

Meters

Example: **2.8 m**

CALCULATE TOTAL FLOW RATE



Remember the formula:

$$\begin{array}{c} \text{TOTAL FLOW RATE} \\ \text{(L/min)} \end{array} \begin{array}{c} \text{APPLICATION VOLUME} \\ \text{(L/ha)} \end{array} \begin{array}{c} \text{FORWARD SPEED} \\ \text{(km/h)} \end{array} \begin{array}{c} \text{WORKING WIDTH} \\ \text{(m)} \end{array} \div \begin{array}{c} \text{600} \\ \text{CONVERSION CONSTANT} \\ \text{(ha and km/h} \\ \text{to L/min)} \end{array}$$

The diagram illustrates the formula for calculating the total sprayer flow rate. On the left, a box labeled 'TOTAL FLOW RATE (L/min)' contains an icon of a measuring jug. This is followed by an equals sign. To the right of the equals sign are three boxes: 'APPLICATION VOLUME (L/ha)' with a grid icon and '1 ha' label, 'FORWARD SPEED (km/h)' with a tractor icon, and 'WORKING WIDTH (m)' with a tree icon and a double-headed arrow labeled 'm'. These three boxes are multiplied together. Finally, the result is divided by a box labeled '600 CONVERSION CONSTANT (ha and km/h to L/min)'.



The result of the formula will then be the total sprayer flow rate in L/min.

 RENOVATE APP

3.5

**CALCULATE NOZZLE
FLOW RATE**

CALCULATE NOZZLE FLOW RATE



So, Laura. Now you are ready to calculate the **nozzle flow rate!!!**



The nozzle flow rate is obtained by dividing the total flow rate by the number of active nozzles (N) of your sprayer (we decided to have 6 in total).

$$Q_{\text{nozzle}} = \frac{Q_{\text{total}}}{N}$$

$$Q_{\text{nozzle}} = \frac{5.83}{6}$$

$$Q_{\text{nozzle}} = 0.97 \text{ L/min}$$



RENOVATE APP



NOZZLE SELECTION



HOW TO USE NOZZLE TABLE

Pressure (bar)	Flow Rate (L/min)											
	Purple 0050	Pink 0075	Orange 01	Green 015	Yellow 02	Violet 025	Blue 03	Red 04	Brown 05	Grey 06	White 08	Black 10
1	-	-	0.23	0.35	0.46	0.58	0.69	0.92	1.15	1.39	1.85	2.31
1.5	-	-	0.28	0.42	0.57	0.71	0.85	1.13	1.41	1.70	2.26	2.83
2	-	-	0.33	0.49	0.65	0.82	0.98	1.31	1.63	1.96	2.61	3.27
2.5	-	-	0.37	0.55	0.73	0.91	1.10	1.46	1.83	2.19	2.92	3.65
3	-	-	0.40	0.60	0.80	1.00	1.20	1.60	2.00	2.40	3.20	4.00
4	-	-	0.46	0.69	0.92	1.15	1.39	1.85	2.31	2.77	3.70	4.62
5	-	0.39	0.52	0.77	1.03	1.29	1.55	2.07	2.58	3.10	4.13	5.16
6	-	0.42	0.57	0.85	1.13	1.41	1.70	2.25	2.83	3.39	4.53	5.66
7	0.31	0.46	0.61	0.92	1.22	1.53	1.83	2.44	3.06	3.67	4.89	6.11
8	0.33	0.49	0.65	0.98	1.31	1.63	1.96	2.61	3.27	3.92	5.23	6.53
9	0.35	0.52	0.69	1.04	1.39	1.73	2.08	2.77	3.47	4.16	5.54	6.93
10	0.37	0.55	0.73	1.10	1.46	1.83	2.19	2.92	3.65	4.38	5.84	7.30
11	0.38	0.57	0.77	1.15	1.53	1.91	2.30	3.06	-	-	-	-
12	0.40	0.60	0.80	1.20	1.60	2.00	2.40	3.20	4.00	4.80	6.40	8.00
13	0.42	0.62	0.83	1.25	1.67	2.08	2.50	3.33	-	-	-	-
14	0.43	0.65	0.86	1.29	1.73	2.16	2.59	3.46	4.32	5.19	7.39	9.24
15	0.45	0.67	0.89	1.34	1.79	2.24	2.68	3.58	-	-	-	-

1. Find the required flow rate (L/min)

2. Look for the closest value in the table.

3. Identify the nozzle colour and pressure.

4. Select the combination that best matches the required flow rate.

HOW TO USE NOZZLE TABLE

Pressure (bar)	Flow Rate (L/min)											
	Purple 0050	Pink 0075	Orange 01	Green 015	Yellow 02	Violet 025	Blue 03	Red 04	Brown 05	Grey 06	White 08	Black 10
1	-	-	0.23	0.35	0.46	0.58	0.69	0.92	1.15	1.39	1.85	2.31
1.5	-	-	0.28	0.42	0.57	0.71	0.85	1.13	1.41	1.70	2.26	2.83
2	-	-	0.33	0.49	0.65	0.82	0.98	1.31	1.63	1.96	2.61	3.27
2.5	-	-	0.37	0.55	0.73	0.91	1.10	1.46	1.83	2.19	2.92	3.65
3	-	-	0.40	0.60	0.80	1.00	1.20	1.60	2.00	2.40	3.20	4.00
4	-	-	0.46	0.69	0.92	1.15	1.39	1.85	2.31	2.77	3.70	4.62
5	-	0.39	0.52	0.77	1.03	1.29	1.55	2.07	2.58	3.10	4.13	5.16
6	-	0.42	0.57	0.85	1.13	1.41	1.70	2.26	2.83	3.39	4.53	5.66
7	0.31	0.46	0.61	0.92	1.22	1.53	1.83	2.44	3.06	3.67	4.89	6.11
8	0.33	0.49	0.65	0.98	1.31	1.63	1.96	2.61	3.27	3.92	5.23	6.53
9	0.35	0.52	0.69	1.04	1.39	1.73	2.08	2.77	3.47	4.16	5.54	6.93
10	0.37	0.55	0.73	1.10	1.46	1.83	2.19	2.92	3.65	4.38	5.84	7.30
11	0.38	0.57	0.77	1.15	1.53	1.91	2.30	3.06	-	-	-	-
12	0.40	0.60	0.80	1.20	1.60	2.00	2.40	3.20	4.00	4.80	6.40	8.00
13	0.42	0.62	0.83	1.25	1.67	2.08	2.50	3.33	-	-	-	-
14	0.43	0.65	0.86	1.29	1.73	2.16	2.59	3.46	4.32	5.19	7.39	9.24
15	0.45	0.67	0.89	1.34	1.79	2.24	2.68	3.58	-	-	-	-



For airblast sprayers, the recommended pressure range is usually between 6 and 12 bar.



Very low pressures may result in poor atomisation and spray distribution, while very **high pressures** produce finer droplets, increasing the **risk of drift**.



The objective is to **achieve the required flow rate using an appropriate nozzle size** and a suitable operating pressure.



NOZZLE SELECTION

Nozzle flow chart – ISO 10625

Pressure (bar)	Flow Rate (L/min)											
	Purple 0050	Pink 0075	Orange 01	Green 015	Yellow 02	Violet 025	Blue 03	Red 04	Brown 05	Gray 06	White 08	Black 10
1	-	-	0.23	0.35	0.46	0.58	0.69	0.97	1.15	1.39	1.85	2.31
1.5	-	-	0.28	0.42	0.57	0.71	0.85	1.14	1.41	1.70	2.26	2.83
2	-	-	0.33	0.49	0.65	0.82	0.98	1.31	1.63	1.96	2.61	3.27
2.5	-	-	0.37	0.55	0.73	0.91	1.10	1.46	1.83	2.19	2.92	3.65
3	-	-	0.40	0.60	0.80	1.00	1.20	1.60	2.00	2.40	3.20	4.00
4	-	-	0.46	0.69	0.92	1.19	1.39	1.85	2.31	2.77	3.70	4.62
5	-	0.39	0.52	0.77	1.03	1.29	1.55	2.07	2.58	3.10	4.13	5.16
6	-	0.42	0.57	0.85	1.13	1.41	1.70	2.26	2.83	3.39	4.53	5.66
7	0.31	0.46	0.61	0.92	1.22	1.53	1.83	2.44	3.06	3.67	4.89	6.11
8	0.33	0.49	0.65	0.98	1.31	1.63	1.96	2.61	3.27	3.92	5.23	6.53
9	0.35	0.52	0.69	1.04	1.39	1.73	2.08	2.77	3.47	4.16	5.54	6.93
10	0.37	0.55	0.73	1.10	1.46	1.83	2.19	2.92	3.65	4.38	5.84	7.30
11	0.38	0.57	0.77	1.15	1.53	1.91	2.30	3.06	-	-	-	-
12	0.40	0.60	0.80	1.20	1.60	2.00	2.40	3.20	4.00	4.80	6.40	8.00
13	0.42	0.62	0.83	1.25	1.67	2.08	2.50	3.33	-	-	-	-
14	0.43	0.65	0.86	1.29	1.73	2.16	2.59	3.46	4.32	5.19	7.39	9.24
15	0.45	0.67	0.89	1.34	1.79	2.24	2.68	3.58	-	-	-	-

Laura needs a nozzle flow rate of approximately 0.97 L/min.

Which nozzle-pressure combination would be the most appropriate for a conventional airblast sprayer?

A

ISO 04 at 1 bar.

B

ISO 025 at 3 bar.

C

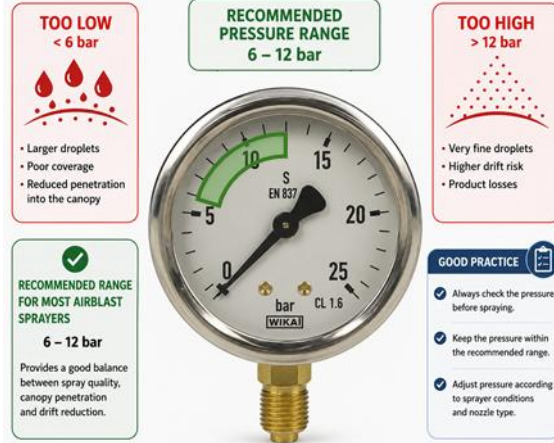
ISO 015 at 8 bar.

D

ISO 01 at 15 bar.



NOZZLE SELECTION



CORRECT

C

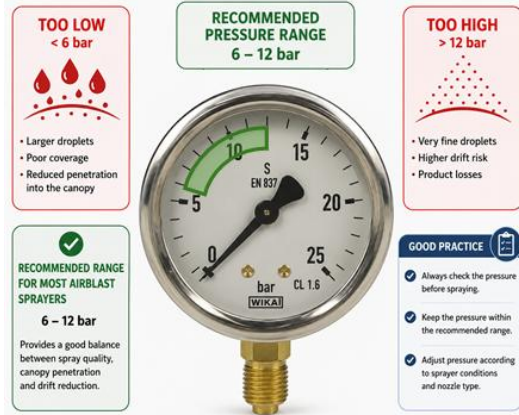
ISO 015 at 8 bar.

i

Different nozzle sizes and pressures can deliver the same flow rate. Choose a pressure within the recommended range for the best results.



NOZZLE SELECTION



INCORRECT

- A** ISO 015
8 bar.
- B** ISO 025
3 bar.
- D** ISO 01
15 bar.

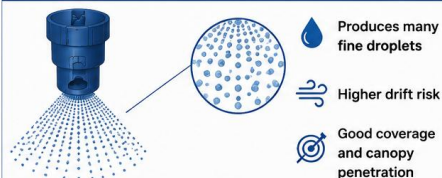
i When selecting a nozzle, you must consider **both the required flow rate and the operating pressure**. Select the pressure within the recommended range for efficient spraying **(6-12 bar)**.

NOZZLE SELECTION

Mia's tip:

- i** Air-induction nozzles produce larger droplets containing air bubbles. These droplets are less likely to drift away from the target and are commonly used when drift reduction is a priority.

CONVENTIONAL HOLLOW-CONE NOZZLE



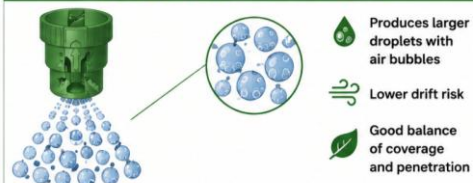
Produces many fine droplets

Higher drift risk

Good coverage and canopy penetration

Best use: Situations where drift is not a major concern and maximum coverage is required.

AIR-INDUCTION (ANTI-DRIFT) NOZZLE



Produces larger droplets with air bubbles

Lower drift risk

Good balance of coverage and penetration

Best use: Situations where drift reduction is important (windy conditions, sensitive areas, regulations).

NOZZLE SELECTION



Laura, remember that for most vineyard fungicide applications **Medium** (M) to **Coarse** (C) droplets often provide a good compromise between:

- ✓ Good coverage
- ✓ Canopy penetration
- ✓ Drift reduction



NOZZLE SELECTION



Laura, which option would be better for reducing spray drift?



A

Conventional nozzle.

B

Air induction nozzle.

C

Higher pressure nozzle.

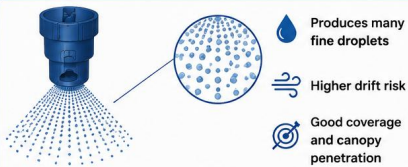
D

Smaller nozzle size.



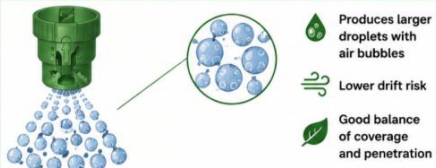
NOZZLE SELECTION

CONVENTIONAL HOLLOW-CONE NOZZLE



Best use: Situations where drift is not a major concern and maximum coverage is required.

AIR-INDUCTION (ANTI-DRIFT) NOZZLE



Best use: Situations where drift reduction is important (windy conditions, sensitive areas, regulations).

CORRECT

B

Air induction nozzle.

i

Air induction nozzles produce larger droplets that are less likely to drift away from the target. They are a good choice for reducing spray drift.



NOZZLE SELECTION

INCORRECT

A

Conventional nozzle.

C

Higher pressure nozzle.

D

Smaller nozzle size.



To reduce spray drift, choose a nozzle that produces larger droplets.

 RENOVATE APP

4.1

**NON-UNIFORM
CANOPY NOZZLES
SELECTION**

NON-UNIFORM CANOPY

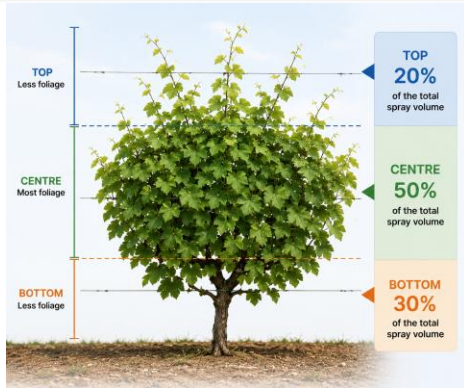


i When the canopy is not uniform, the total flow rate can be distributed according to the vegetation density. Different nozzle sizes may be used in each section of the sprayer, but the nozzle configuration should be symmetrical on both sides of the sprayer, and all nozzles must operate at the same pressure.



NON-UNIFORM CANOPY

In case of heterogeneous canopy, what is the best calibration strategy?



- A** Same nozzle in all position.
- B** Different nozzle sizes at different positions.
- C** Asymmetric nozzle configuration.
- D** Different pressure at different position.



NON-UNIFORM CANOPY



CORRECT

B

Different nozzle sizes at different positions.

i

When the canopy is not uniform, the spray volume should be adjusted to the vegetation density. This is achieved by using **different nozzle sizes** in each section while maintaining the **same operating pressure** throughout the sprayer.



NON-UNIFORM CANOPY



INCORRECT

A Same nozzle in all position.

C Asymmetric nozzle configuration.

D Different pressure at different positions.

i In a heterogeneous canopy, the spray volume should match the vegetation density. The best solution is to use **different nozzle sizes** for each section while keeping the **same pressure** across the entire sprayer.

 RENOVATE APP

5

VERIFICATION OF
THE RESULTS

VERIFICATION OF THE RESULTS

Great! Now it's time to verify that our sprayer is working correctly. We will check **four important aspects** before we start spraying.

WE WILL CHECK:

- 1 CHECK NOZZLE FLOW RATE**
We verify that each nozzle is delivering the correct flow rate according to the selected pressure. This ensures that the amount of product applied per hectare is accurate and uniform across the whole field.
- 2 VERIFICATION OF AIR FLOW DIRECTION**
We check that the air is directed correctly towards the canopy and reaches the target area. Proper air direction is essential for good penetration and to ensure that the product reaches all parts of the plants.
- 3 VERIFICATION OF AIR FLOW VOLUME**
We make sure the air volume is adequate for good penetration. The correct airflow volume allows the product to cover the canopy efficiently without causing drift or product losses.
- 4 SPRAY COVERAGE TEST WITH WSP**
We use water sensitive paper to evaluate the quality of the spray coverage on the leaves. This test helps us confirm that the droplets are well distributed across the canopy for an effective application.

 RENOVATE APP

5.1

**CHECK NOZZLE
FLOW RATE**

ALWAYS CHECK NOZZLE FLOW RATE



Laura, the nozzle chart provides a theoretical flow rate. Always verify the actual nozzle output by measuring it in the field before spraying.

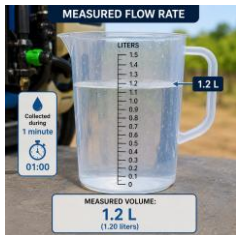


i Nozzle flow rate must be checked periodically. Nozzles must be replaced when the discharged flow rate differs more than 10% from the nominal flow rate (ISO 16122).



ALWAYS CHECK NOZZLE FLOW RATE

Laura measures the flow rate of one nozzle and finds out it doesn't match the expected one. What do you think to be the cause of it?



Required:
0.97 L/min

Measured:
1.20 L/min

A

Worn nozzle.

B

Damaged manometer.

C

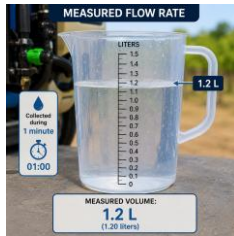
Too high pressure.

D

Large row spacing.



ALWAYS CHECK NOZZLE FLOW RATE



CORRECT

A

Worn nozzle.

C

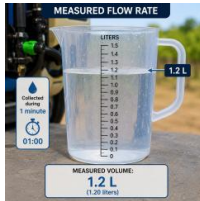
Too high pressure.



Excessive nozzle flow is usually caused by **nozzle wear**, **high pressure**, or an **incorrect pressure reading**. Check the equipment before spraying.



ALWAYS CHECK NOZZLE FLOW RATE



INCORRECT

B

Damaged manometer.

D

Large row spacing.

i

When the measured flow rate is much higher than expected the most likely cause is a **worn nozzle**. Also check that the pressure is correct and the manometer is not damaged. Checking nozzle flow regularly helps maintain the correct application rate.

 RENOVATE APP

5.2

**VERIFICATION OF AIR
FLOW DIRECTION**

RENOVATE APP



MIA'S TIP

The ribbons should reach the canopy, but they should not stay above it or fall below it. Air must go through the canopy where the leaves are.

WHAT ARE WE LOOKING FOR?



AIR REACHES THE WHOLE CANOPY

The ribbons should reach the leaves throughout the entire height and depth of the canopy.



AIR IS DIRECTED TOWARDS THE LEAVES

The airflow should be aimed into the canopy, not upwards or downwards. This ensures good penetration.



NO STRONG AIR ESCAPING ABOVE OR BELOW THE CANOPY

Ribbons should not lift up over the top of the canopy or drop under it. Excess air in those areas reduces efficiency.



WHY IS THIS IMPORTANT?

Correct air direction ensures the product reaches the target area, improves coverage and effectiveness, and avoids waste.



VERIFICATION OF AIR FLOW DIRECTION

What is the purpose of using
ribbons to check air flow
directions?

A

To measure tractor speed.

B

To verify that the air is directed
towards the canopy.

C

To measure spray pressure.

D

To estimate tank capacity.



VERIFICATION OF AIR FLOW DIRECTION

What is the purpose of using
ribbons to check air flow
direction?

CORRECT

B

To verify that the air is directed
towards the canopy.



The ribbons make the air movement
visible, allowing you to verify that the
airflow is correctly directed into the
canopy.



VERIFICATION OF AIR FLOW DIRECTION

What is the purpose of using
ribbons to check air flow
direction?

INCORRECT

A

To measure tractor speed.

C

To measure spray pressure.

D

To estimate tank capacity.



The ribbons do not measure speed, pressure or tank capacity. They make the air movement visible so you can verify that the airflow is correctly directed towards the canopy.



VERIFICATION OF AIR FLOW DIRECTION

If the ribbons point well above
the canopy, what does it
indicate?

A The airflow direction is correct.

B The nozzles are blocked.

C The airflow is poorly adjusted and
should be redirected.

D The spray pressure is too low.



VERIFICATION OF AIR FLOW DIRECTION

If the ribbons point well above
the canopy, what does it
indicate?

CORRECT

C

The airflow is poorly adjusted and
should be redirected.

i

Air should be directed towards the
vegetation. Air flowing above the
canopy carries droplets away from the
target.



VERIFICATION OF AIR FLOW DIRECTION

If the ribbons point well above
the canopy, what does it
indicate?

INCORRECT

A The airflow direction is correct.

B The nozzles are blocked.

D The spray pressure is too low.

i Air should be directed into the canopy,
not above it. If the ribbons point above
the vegetation, the airflow direction is
not correctly adjusted and droplets may
miss the target, increasing drift.

 RENOVATE APP

5.3

**VERIFICATION OF AIR
FLOW VOLUME**

RENOVATE APP



Now we check the air flow volume.
Too much air can push spray through the canopy and increase losses.
The goal is a properly adjusted air flow.

METHODS TO ADJUST AIR FLOW VOLUME

1. SELECTION OF GEAR AND RPM OF TRACTOR ENGINE



Adjust the gear and engine RPM to change the fan speed.

2. SELECTION OF FASTER OR SLOWER GEAR ON FAN TRANSMISSION OF THE SPRAYER



Use the lever to select a faster or slower gear on the fan transmission.

3. CHANGING FAN BLADE ANGLE



Change the angle of the fan blades to increase or decrease the air volume.

IMPROPER AIR FLOW SETTING – TOO STRONG AIR FLOW



CORRECT AIR FLOW SETTING – PROPERLY ADJUSTED AIR FLOW



MIA'S TIP

Air flow volume must be balanced.
The right setting improves penetration and coverage, while reducing drift and product losses.



VERIFICATION OF AIR FLOW VOLUME

What happens if the air flow
volume is too high?

A

Coverage always improves.

B

Spray may pass through the canopy
and reach the next row.

C

The nozzles stop spraying.

D

Water consumption decreases.



VERIFICATION OF AIR FLOW DIRECTION

What happens if the air flow
volume is too high?

CORRECT

B

Spray may pass through the
canopy and reach the next row.

i

Excessive air pushes droplets through
the canopy, increasing spray losses and
reducing application efficiency.



VERIFICATION OF AIR FLOW DIRECTION

What happens if the air flow
volume is too high?

INCORRECT

A Coverage always improves.

C The nozzles stop spraying.

D Water consumption decreases.

i Excessive air flow pushes droplets through the canopy and into the next row instead of depositing them on the leaves. The objective is to use only the amount of air needed to penetrate the canopy.

 RENOIVATE APP

5.4

SPRAY COVERAGE

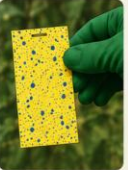
TEST WITH WSP



HOW TO PERFORM THE SPRAY COVERAGE TEST

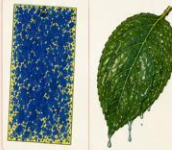
- 1 PLACE THE WSP ON THE LEAVES**

 Attach Water Sensitive Papers on both sides of the leaves at different heights and positions within the canopy.
- 2 APPLY THE SPRAY AS USUAL**

 Make your application using the normal operating settings.
- 3 COLLECT AND OBSERVE THE WSP**

 After spraying, collect the WSP and let them dry. Observe the number and size of the droplets.
- 4 EVALUATE THE COVERAGE**

 Compare the coverage with the examples below to determine if the application is adequate.

INTERPRETATION OF SPRAY COVERAGE

- | VERY POOR COVERAGE | SUFFICIENT COVERAGE | EXCESSIVE COVERAGE |
|--|--|--|
|  |  |  |
| <p>✗ Not enough droplets on the leaf. Poor coverage reduces efficacy.</p> | <p>✓ Good number and size of droplets. The leaf is well covered.</p> | <p>✗ Too many droplets and runoff. Risk of losses and lower efficiency.</p> |



MIA'S TIP

Using WSP is a simple and effective way to check your application quality. A good coverage improves efficacy, reduces waste and protects the environment.



SPRAY COVERAGE TEST WITH WSP

What is the purpose of Water
Sensitive Paper (WSP)?

A

To check relative humidity.

B

To measure the spray pressure.

C

To measure the wind speed.

D

To evaluate how well the spray
covers the target.



SPRAY COVERAGE TEST WITH WSP

What is the purpose of Water Sensitive Paper (WSP)?

CORRECT

D

To evaluate how well the spray covers the target.

i

Water Sensitive Papers show where droplets reach the canopy and help assess spray coverage quantity.



SPRAY COVERAGE TEST WITH WSP

What is the purpose of Water Sensitive Papers (WSP)?

INCORRECT

A To check relative humidity.

B To measure the spray pressure.

C To measure the wind speed.

i Water Sensitive Papers are not used to measure speed, wind, or pressure. They are used to evaluate how well the spray reaches the target and whether the droplet coverage is adequate.